

● MEDIUM

● **ADVANCED MATH**

Advanced Math

10 questions · ~15 min

11

Q1

ADVANCED MATH

$$\frac{2(x+1)}{x+5} = 1 - \frac{1}{x+5}$$

What is the solution to the equation above?

- A. 0
- B. 2
- C. 3
- D. 5

Q2

ADVANCED MATH

Which of the following is equivalent to the expression $x^4 - x^2 - 6$?

- A. $(x^2 + 1)(x^2 - 6)$
- B. $(x^2 + 2)(x^2 - 3)$
- C. $(x^2 + 3)(x^2 - 2)$
- D. $(x^2 + 6)(x^2 - 1)$

Q3

ADVANCED MATH

$$px + 57 = x^2$$

The given equation relates the value of x and its corresponding value of px for the function p . What is the minimum value of the function p ?

- A. -3,249
- B. -57
- C. 57
- D. 3,249

Q4

ADVANCED MATH

$$y = ax^2 - c$$

In the equation above, a and c are positive constants. How many times does the graph of the equation above intersect the graph of the equation $y = a + c$ in the xy -plane?

- A. Zero
- B. One
- C. Two
- D. More than two

Q5

GRID-IN

ADVANCED MATH

$$2x^2 + 5x - 12$$

If the given expression is rewritten in the form $(2x - 3)(x + k)$, where k is a constant, what is the value of k ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

ADVANCED MATH

A rectangle has a length that is 15 times its width. The function $y = 15ww$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. Which of the following is the best interpretation of $15w$ in this context?

- A. The length of the rectangle, in feet
- B. The area of the rectangle, in square feet
- C. The difference between the length and the width of the rectangle, in feet
- D. The width of the rectangle, in feet

Q7

ADVANCED MATH

$$(x - 4)(x + 2)(x - 1) = 0$$

What is the product of the solutions to the given equation?

- A. 8
- B. 3
- C. -3
- D. -8

Q8

ADVANCED MATH

Which expression is equivalent to $\frac{h^{15}q^7}{h^5q^{21}}$, where $h > 0$ and $q > 0$?

- A. $\frac{h^{10}}{q^{14}}$
- B. $\frac{h^3}{q^3}$
- C. $h^{10}q^{14}$
- D. h^3q^3

Q9

ADVANCED MATH

$$f(x) = x^3 + 3x^2 - 6x - 1$$

For the function f defined above, what is the value of $f(-1)$?

- A. -11
- B. -7
- C. 7
- D. 11

$$7m = 5n + p$$

The given equation relates the positive numbers m , n , and p . Which equation correctly gives n in terms of m and p ?

A. $n = \frac{5p}{7m}$

B. $n = \frac{7m}{5} - p$

C. $n = 57m + p$

D. $n = 7m - 5 - p$